

Week 3 – Designing the Database Structure

Title

Entity Relationship (ER) Diagram and Relational Schema Design

Objective

The objective of this phase is to design the database structure for the **E-Commerce Order Management System**. This includes identifying entities, defining relationships between them, creating an Entity Relationship (ER) Diagram, and converting the ER model into a relational schema that will later be implemented in SQL.

Entity Identification

The following entities are identified:

Entity	Description
Customer	Stores customer information
Product	Stores product details
Category	Stores product categories
Supplier	Stores supplier information
Orders	Stores customer orders
Order_Item	Stores products included in each order
Payment	Stores payment details
Shipment	Stores shipping information

Attributes of Each Entity

1. Customer

- Customer_ID (PK)
 - Name
 - Email
 - Phone
 - Address
-

2. Category

- Category_ID (PK)
 - Category_Name
-

3. Supplier

- Supplier_ID (PK)
 - Supplier_Name
 - Contact_Number
 - Address
-

4. Product

- Product_ID (PK)
 - Product_Name
 - Price
 - Stock
 - Category_ID (FK)
 - Supplier_ID (FK)
-

5. Orders

- Order_ID (PK)
 - Order_Date
 - Status
 - Customer_ID (FK)
-

6. Order_Item

- Order_Item_ID (PK)
 - Quantity
 - Unit_Price
 - Order_ID (FK)
 - Product_ID (FK)
-

7. Payment

- Payment_ID (PK)
- Payment_Method
- Payment_Status

- Payment_Date
 - Order_ID (FK)
-

8. Shipment

- Shipment_ID (PK)
 - Shipment_Date
 - Delivery_Date
 - Shipment_Status
 - Order_ID (FK)
-

Relationship Identification

1. Customer → Orders

Relationship:

One Customer can place many Orders

Type: One-to-Many (1:M)

2. Category → Product

Relationship:

One Category contains many Products

Type: One-to-Many (1:M)

3. Supplier → Product

Relationship:

One Supplier supplies many Products

Type: One-to-Many (1:M)

4. Orders → Order_Item

Relationship:

One Order contains many Order Items

Type: One-to-Many (1:M)

5. Product → Order_Item

Relationship:

One Product can appear in many Order Items

Type: One-to-Many (1:M)

6. Orders → Payment

Relationship:

Each Order has one Payment

Type: One-to-One (1:1)

7. Orders → Shipment

Relationship:

Each Order has one Shipment

Type: One-to-One (1:1)

Many-to-Many Relationship

Customers purchase multiple products.

Products are purchased by multiple customers.

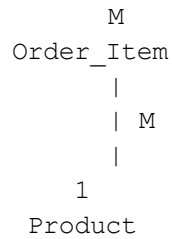
Therefore,

Customer ↔ Product

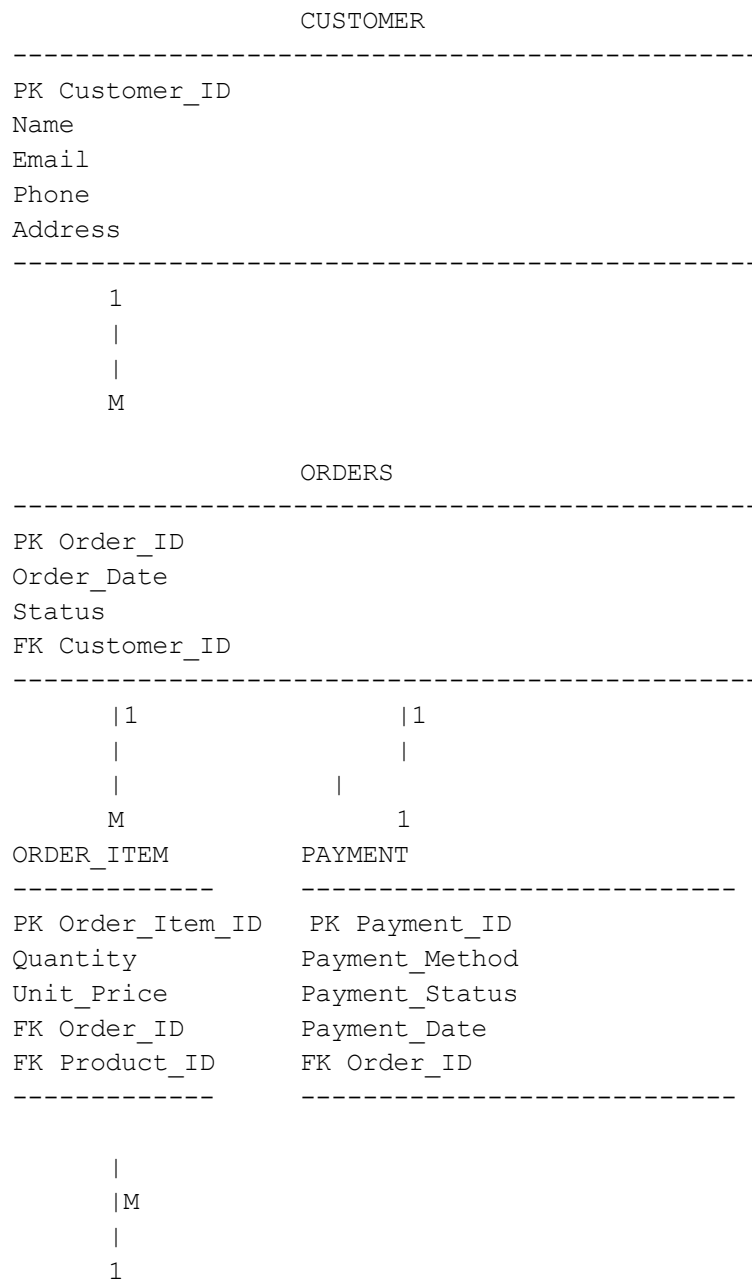
is a **Many-to-Many (M:N)** relationship.

This relationship is resolved using the **Order_Item** table.

```
Customer
  |
  | 1
  |
  M
Orders
  |
  | 1
  |
```



Entity Relationship Diagram (ERD)



PRODUCT	

PK Product_ID	
Product_Name	
Price	
Stock	
FK Category_ID	
FK Supplier_ID	

1	1
M	M
CATEGORY	SUPPLIER
-----	-----
PK Category_ID	PK Supplier_ID
Category_Name	Supplier_Name
	Contact_Number
	Address
-----	-----

ORDERS

|
|1
|
1

SHIPMENT

PK Shipment_ID
Shipment_Date
Delivery_Date
Shipment_Status
FK Order_ID

Relational Schema

CUSTOMER (
Customer_ID PK,
Name,
Email,
Phone,
Address

```
)

CATEGORY (
    Category_ID PK,
    Category_Name
)

SUPPLIER (
    Supplier_ID PK,
    Supplier_Name,
    Contact_Number,
    Address
)

PRODUCT (
    Product_ID PK,
    Product_Name,
    Price,
    Stock,
    Category_ID FK,
    Supplier_ID FK
)

ORDERS (
    Order_ID PK,
    Order_Date,
    Status,
    Customer_ID FK
)

ORDER_ITEM (
    Order_Item_ID PK,
    Quantity,
    Unit_Price,
    Order_ID FK,
    Product_ID FK
)

PAYMENT (
    Payment_ID PK,
    Payment_Method,
    Payment_Status,
    Payment_Date,
    Order_ID FK
)

SHIPMENT (
    Shipment_ID PK,
    Shipment_Date,
```

```
        Delivery_Date,  
        Shipment_Status,  
        Order_ID FK  
    )
```

Primary Keys (PK)

- Customer_ID
 - Category_ID
 - Supplier_ID
 - Product_ID
 - Order_ID
 - Order_Item_ID
 - Payment_ID
 - Shipment_ID
-

Foreign Keys (FK)

- Customer_ID → Orders
 - Category_ID → Product
 - Supplier_ID → Product
 - Order_ID → Order_Item
 - Product_ID → Order_Item
 - Order_ID → Payment
 - Order_ID → Shipment
-

Conclusion

The ER diagram identifies the entities, attributes, primary keys, foreign keys, and relationships required for the E-Commerce Order Management System. The many-to-many relationship between customers and products is resolved using the **Order_Item** entity. The relational schema provides a structured blueprint for implementing the database in SQL during the next phase of the project.